

FPV RC Arena — Build Plan v1

Customers drive real 1:14–1:16 RC cars from a racing-wheel cockpit and watch the car's live 5.8 GHz camera feed on a monitor or in goggles. Prices and stock verified from Indian sellers on 9 Sep 2026 unless marked **verify**. Cars, batteries, chargers and car spares are owned and excluded from every total. Dry-run findings that shaped this version: [FPV RC Arena - Dry Run Report.md](#).

1. Scope

Item	Decision
Opening configuration	2 stations: 1 × Wheel + Monitor, 1 × Wheel + Goggles (goggles lane is staff-attended)
Cars in service	2 per station (one running, one on charge) → 4 fitted cars + 1 spare camera cartridge
Car class	1:14–1:16 off-road, 2S; single model, locked before ordering (section 5)
Track	Indoor, barrier-enclosed, barriers under 300 mm so the marshal can reach in
Video	Analog 5.8 GHz, 25 mW, Raceband R3 + R6 (R4/R5 reserved for stations 3–4). Camera + VTX on their own 2S pack — never the car pack
Customer display	Monitor lane: 27-inch gaming monitor fed from the 7-inch FPV receiver's AV out. Goggles lane: Skyzone Cobra SD. The 7-inch unit is the marshal screen on both lanes
Control	2.4 GHz ExpressLRS, one link per station, unique bind phrase per station
Staff	2 on floor: track marshal + goggles attendant
Build order	Monitor station first → 15-point gate (section 9) → goggles station from the same loom

2. Station architecture

Control: G29 wheel + pedals → USB → Raspberry Pi 5 → CRSF (half-duplex, 400 k baud) over USB-UART → ELRS 2.4 GHz TX module → RadioMaster ER4 receiver in the car → CH1 steering servo · CH2 ESC.

Video: dedicated 2S video pack → 5 V 2 A UBEC → SpeedyBee TX800 VTX (25 mW, pit when parked) + RunCam Phoenix 2 camera → MMCX pigtail → RHCP lollipop on the rear bulkhead → 5.8 GHz → 2 × RHCP antennas → Eachine LCD5802D 7-inch diversity receiver/DVR (marshal screen) → AV out → AV-to-HDMI upscaler → **27-inch monitor** at the cockpit (monitor lane) or Skyzone Cobra SD goggles on their own receiver (goggles lane).

Element	Spec
Pi software	Reads wheel/pedals via <code>evdev</code> ; applies speed level, steering rate limit, session timer, stop logic; streams CRSF at 150 Hz; shows link quality from ELRS telemetry; staff web UI. Read-only root filesystem. One codebase for all stations.
G29	PS3-mode switch; light centring spring only (no game FFB). No wheel or pedal set is sold separately in India — a spare is a whole unit.
CRSF wiring	TX module's CRSF pin is one-wire half-duplex: adapter RXD straight to the pin, TXD through 1 k Ω . Use CP2102N or FT232RL (both do 400 k; the older CP2102 has baud error at 400 k — verify on bench).
ELRS TX module	Powered from the station 5 V rail through the E-stop chain; model-match on; 100 mW; 150 Hz; unique bind phrase per station. Buy all modules in one order — Indian stock is thin.
ER4 receiver	Failsafe per output in its web UI: steering 1500 μ s, throttle 1500 μ s (neutral). Never a brake value on a brushed ESC — below-neutral after neutral means reverse. Enable ESC drag brake where offered.
Video pack	2S 450–1000 mAh XT30, velcro'd low in the chassis, with a LiPo buzzer alarm. Electrically isolated from the car pack: no motor/ESC noise on the video, no brown-out under throttle, no LC filter. ~ 0.5 A at 5 V $\rightarrow$ 450 mAh $\approx$ 1 h, 850 mAh $\approx$ 2 h. Swap on every 4th car-battery swap.
Display	Monitor lane: LG 27GS60F 27-inch 1080p IPS on a VESA-100 arm clamped to the cockpit, fed by the LCD5802D's AV out through an AV-to-HDMI upscaler. Analog 480-line video looks soft at 27 inch; the upscaler adds 1–3 frames — bench-measure glass-to-glass latency; if over 120 ms, the 7-inch goes back on the cockpit and the 27-inch becomes the spectator screen.
Camera pod	Camera behind the windscreen line in a TPU 95A cradle with lens hood; 2 mm spring-steel wire hoop bonded into the TPU as the roll bar; VTX low under the shell on foam; antenna on a rear bulkhead SMA via the pigtail so it is never a lever on the VTX. Cartridge = camera + cradle + hoop + keyed plug, 2 screws, swap under 3 min.
Cockpit	Gass Power GT Lite, G29 on the deck, 27-inch on a VESA arm (monitor lane); 7-inch receiver on a $\frac{1}{4}$ -inch clamp arm at the marshal's side. 12 V 5 A SMPS per station feeds the 7-inch and the 5 V rails; the 27-inch on its own mains adaptor.
Car ID	LED strip on each car in its station's colour.

3. Safety system

Layer	Trigger	Action	Latency
Software stop (primary)	Station stop button; G29 unplugged; timer end; staff UI	Brake pulse only if the car was moving forward, else neutral; hold neutral until staff clears	under 50 ms
Heartbeat watchdog	Pi GPIO heartbeat stops (crash, hang, power loss)	Missing-pulse detector drops the relay feeding the TX module's 5 V $\rightarrow$ receiver failsafe	under 1.5 s
Hardware E-stop	22 mm mushroom on the cockpit, NC contact in series with the TX module's 5 V	Module dies $\rightarrow$ receiver failsafe (centre, neutral)	up to 1 s

Layer	Trigger	Action	Latency
Marshal master stop	2-NC mushroom in series with both stations' TX 5 V feeds	Both cars failsafe together	up to 1 s
Goggles pendant	Second mushroom on a 2 m lead at the attendant's standing position	Same as station E-stop	up to 1 s

- **Speed caps are a safety control.** The hardware path has up to 1 s dead time (5.5 m at 20 km/h). Throttle caps **40 % child / 60 % adult / 100 % staff** + steering rate limit, set by staff per session, not changeable from the wheel. Last 20 s of every session drop to 20 % "return to pit".
- **Every (re)connect starts disarmed** at neutral until staff clears — after an E-stop, a watchdog trip, or a Pi reboot.
- **Marshal view is independent** of the customer display on both lanes: the 7-inch LCD5802D is the marshal/attendant screen (own receiver, same channel). Audible "car lost link" alert from telemetry.
- **Loss of video ≠ loss of control.** Tested separately (gate item 8).

4. RF plan

- **5.8 GHz video.** Legal band 5725–5875 MHz (GSR 1048(E), 2018). Cars lock to **R3 5732 MHz** and **R6 5843 MHz** at 25 mW (R4/R5 for expansion). R1, R2, R7, R8 and Band E are outside the band and never used. RHCP both ends. Channel lock documented per VTX.
- **Venue Wi-Fi.** 5 GHz channels 149–165 overlap the video channels. Written agreement: access points near the arena on 2.4 GHz or 5 GHz channels 36–64.
- **2.4 GHz control.** ELRS at 100 mW is inside GSR 45(E) 2005. No constraint.

5. Car platform (lock before ordering)

All candidate cars run 2S packs that sag to ~6.5 V under brushed-motor load — this is why the camera and VTX get their own pack.

Car	Electronics	Work per car
MJX Hyper Go 16208 / 16207 / 14301 / 14302	Discrete receiver, 3-wire servo	ER4 swap only — recommended prototype car
WLtoys 144010 / 124017 / 124008 (brushless)	Discrete receiver	ER4 swap only
Traxxas 1/16	Discrete receiver	ER4 swap only
WLtoys 144001 / 124018 / 124019 (brushed)	Integrated RX+ESC board (1311), proprietary 5-wire servo (1307)	New brushed ESC + 3-wire servo + ER4 ≈ ₹2,800/car (section 6.3)
HBX 16889	Integrated ESC+RX, standard servo	New brushed ESC + ER4

6. Bill of materials

6.1 Station base (both stations)

Item	Product	Seller	Price
Wheel + pedals	Logitech G29	Flipkart ₹29,920 offer / PrimeABGB ₹33,134. Alt with magnetic pedals: Thrustmaster T248 Flipkart ₹33,989	₹30,000–33,500
Cockpit with seat	Gass Power GT Lite	gasspower.in	₹28,999
7-inch mount	¼-inch clamp arm on the frame	generic	₹500–1,000
7-inch diversity receiver + DVR (marshal screen)	Eachine LCD5802D (7–16 V in, 2 × AV out)	RC Mumbai ₹9,249 / rcproduct.in ₹10,999 + tax	₹9,250–11,000
RX antennas ×2	Skyzone mushroom RHCP + Blackberry Lollipop V4 RHCP	hitechxyz ₹549 + hitechxyz ₹349	₹900–1,100
Control computer	Raspberry Pi 5 4 GB + 27 W PSU + 32 GB A2 SD + case/fan + CP2102N or FT232RL USB-UART	ThinkRobotics Pi 5 ₹8,999 · Robu PSU ₹1,255 · ThinkRobotics SD ₹1,480 · USB-UART ₹120–400 verify	₹12,000–14,000
ELRS 2.4 GHz TX module	DakeFPV ELRS Micro TX (in stock) · RadioMaster Ranger Nano (OOS) · Happymodel ES24TX Pro (Amazon, worst case)	hitechxyz ₹5,949 · RC Mumbai ₹4,152 · Amazon ₹9,360	₹4,200–6,000 (₹9,360 worst)
E-stop, watchdog, power, enclosure, labour	LAY37 22 mm mushroom 1NO+1NC · 5 V relay + missing-pulse detector · 12 V 5 A SMPS · 5 V DC-DC · IP65 box · looms · one fabricator day	Robu E-stop ₹99 · relay ₹81 · SMPS ₹395 + GST	₹8,000–15,000
Base subtotal			₹94,000–1,14,000

6.2 Car kit (one per fitted car — 4 cars + 1 spare cartridge)

Item	Product	Seller	Price
Receiver	RadioMaster ER4 ELRS PWM	RC Mumbai	₹2,249
Camera	RunCam Phoenix 2 SE V2 (or SPV5 ₹2,499)	hitechxyz	₹2,500–2,900
VTX	SpeedyBee TX800 (pit / 25 / 200 / 400 / 800 mW, 5 V in, MMCX)	hitechxyz / RC Mumbai	₹3,998
Video pack ×2 (one on the car, one charging)	2S 450–1000 mAh XT30 + LiPo buzzer alarm + XT30 charge lead	hitechxyz / quadkart / Robu Orange verify (small 2S packs are thin in Indian stock — buy all at once)	₹1,300–2,100

Item	Product	Seller	Price
Video power	5 V 2 A UBEC (2–6S in)	Robu / FlyRobo verify	₹250–500
Antenna + pigtail	Blackberry Lollipop V4 RHCP SMA + MMCX–RP–SMA pigtail	hitechxyz ₹349 + pigtail ₹249	₹600
Camera cartridge	TPU 95A cradle + hood, 2 mm spring–steel hoop, keyed plug, silicone wire, breakaway plate	MNM3D quote + Robu	₹1,000–2,500
Per-car subtotal			₹11,900–14,800

6.3 Conditional conversion (car cost, excluded) — brushed WLtoys / HBX only

Brushed 60 A ESC with BEC + failsafe [FlyRobo](#) ₹1,363 · 25 kg metal-gear servo [FlyRobo](#) ₹1,619 / Robu DS3225 ₹1,409 → ≈ **₹2,800 per car** (HBX keeps its servo).

6.4a Monitor station — additions over the base

Item	Product	Seller	Price
27-inch monitor	LG UltraGear 27GS60F 27-inch 1080p IPS, HDMI, VESA 100 (any 24–27-inch 1080p VESA monitor works)	Computech Store ₹12,289 in stock · PrimeABGB OOS · Flipkart list ₹28,500	₹12,000–14,000
AV-to-HDMI upscaler	Mini CVBS → HDMI 1080p (LCD5802D AV out → monitor)	Flipkart ₹999–1,260 / IndianRoboStore	₹500–1,300
VESA arm	Clamp-mount VESA-100 monitor arm on the cockpit frame	generic	₹1,500–3,000
Monitor delta			₹14,000–18,300

6.4b Goggles station — additions over the base

Item	Product	Seller	Price
Customer goggles	Skyzone Cobra SD (diversity, DVR, AV out) — budget alt LS-800D ₹11,499	hitechxyz / LS-800D	₹11,500–23,500
Goggle antennas ×2	RHCP mushroom + lollipop (replace the linear whips)	hitechxyz	₹900
Hygiene	Disposable eye masks 200-pk + 2 spare foam faceplates + wipes + lanyard	Amazon ₹1,699 + foam ₹1,550–2,490 each	₹5,000–7,000
Pendant stop + signage	Mushroom on 2 m lead; motion-sickness / epilepsy / age-height notice	Robu ₹99 + enclosure + print	₹1,500–3,000
Goggles delta			₹19,000–34,500

On both lanes the base 7-inch unit is the marshal/attendant screen — no second receiver.

6.5 Programme items (one-time)

Item	Estimate
Station software (in-house ₹0 / outsourced)	₹0–60,000
Spares: 1 camera cartridge (complete), 2 ER4, 2 antennas + pigtails, 1 ELRS TX module, 1 UBEC, 2 video packs, 1 LCD5802D, 1 upscaler; optional spare G29 ₹33k	₹30,000–63,000
Prototype consumables (50 impacts, 10 rolls, 3 cradle revisions)	₹8,000–12,000
Compliance query (WPC / ETA consultant)	₹0–10,000
Pit safety: LiPo bags, sand bucket, extinguisher	₹3,000–6,000
Marshal master stop (2-NC mushroom) + grabber tool	₹1,000–2,000

7. Budget (cars, batteries, chargers, car spares excluded)

Scope	Range
Wheel + Monitor station (base + 2 car kits + 27-inch delta)	₹1.32 L – 1.62 L
Wheel + Goggles station (base + 2 car kits + goggles delta)	₹1.37 L – 1.78 L
Two stations	₹2.69 L – 3.40 L
Programme total incl. software, spares, prototype, compliance, pit safety	₹3.1 L – 4.9 L
Later: each extra monitor station	+₹1.32–1.62 L

Gate item 14: prototype station actual spend within 15 % of the single-station range.

8. Compliance and paperwork

- **5.8 GHz.** 5725–5875 MHz delicensed under GSR 1048(E) 2018 (secondary sources: 1 W, ≤ 6 dBi). Quote the gazette itself in any venue submission.
- **Analog video under WAS/RLAN wording is a grey area.** One written query to WPC or an ETA consultant before bulk purchase; answer filed (gate item 13).
- **ETA.** Importer's self-declaration (₹10,000/model); no end-user obligation found. Keep vendor invoices per VTX; ask the seller whether the SKU came in under ETA; describe as a ground-vehicle video transmitter, not a drone part.
- **2.4 GHz** ELRS 100 mW: GSR 45(E) 2005 — clear. **Drone Rules 2021** do not apply.
- **Venue:** public-liability cover, signed waiver, age/height rule, motion-sickness/epilepsy notice, Wi-Fi channel agreement in writing.

Sourcing: rcmumbai.com, hitechxyz.in, quadkart.in, electropi.in, robu.in, flyrobo.in, thinkrobotics.com, gasspower.in. quadkopters.com closed; rcbazaar.com broken. Amazon.in / mdcomputers.in / logtech.in

prices unverified until opened by hand.

9. Build sequence and gate

1. Lock the fleet model; fit ER4 (or convert) on two cars.
2. Build the Wheel + Monitor station (6.1 + 2 × 6.2 + 6.4a) and write the station software.
3. Run the gate. **Go only when all 15 pass.**
4. Build the goggles station from the prototype's exact loom + 6.4b. Build the spares kit.
5. Soft launch: timed solo laps; head-to-head only once both feeds and links are stable together.

#	Test	Pass
1	Fleet decision	Model chosen; ER4 (or conversion) proven on one car
2	Control bench	60 min smooth steering/throttle, no uncommanded movement, input latency measured under 50 ms
3	Software stop	Stop button, G29 unplug, timer end: car stops within 100 ms, 20/20
4	Watchdog	Kill the Pi process / pull Pi power: TX module drops, car in failsafe within 1.5 s, 10/10
5	Hardware E-stop	Station mushroom, pendant, master stop (both cars at once): failsafe within 1.5 s, 20/20
6	Speed levels	Child / adult / staff caps and steering rate limit hold under full pedal; return-to-pit mode works
7	Video links	Both VTX on R3/R6 at 25 mW, venue Wi-Fi on: 27-inch, 7-inch and goggles clean over the whole track; no lines at full ESC load; a video pack lasts its full swap schedule
7a	Display latency	Camera-to-27-inch glass-to-glass under 120 ms (phone slow-mo against a stopwatch); else the 7-inch returns to the cockpit
8	Video vs control	Cover RX antenna → control still works. Kill TX module → failsafe. Logged separately
9	Impact	50 barrier hits + 10 rolls: no lens contact, no loose camera, no broken wire, antenna intact, VTX under 70 °C after 20 min
10	Cartridge swap	Under 3 min by staff who did not build it
11	Reset time	30 observed sessions, reset under 3 min incl. battery swap; goggles lane incl. mask change
12	Goggles lane	Attendant sees the same car on the station monitor throughout; pendant reachable; no session lost to fog/dirt
13	Compliance	VTX invoices filed; channel lock per VTX documented; WPC/ETA query sent and answer filed
14	Cost	Prototype spend within 15 % of section 7

10. Open decisions

Decision	Owner	Blocks
Fleet car model (recommend MJX Hyper Go 16208: ER4 swap only)	Yash	Every car kit order
Station software: in-house or outsourced	Yash	Gate items 2–6
WPC / ETA written query	Yash	Bulk VTX purchase, venue submission
ELRS TX module source (3 units: 2 + spare)	Yash	Control chain
Cockpit for later stations: Gass Power vs local MS-tube fabricator	after prototype	Stations 3–4